

1. Let V be a real inner product space. A set $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\} \subseteq V$ is called *orthogonal* if it doesn't contain the zero vector, and if

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0 \text{ for all } i \neq j.$$

(a) Let $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be an orthogonal set and suppose that

$$\mathbf{v} = a_1 \mathbf{u}_1 + a_2 \mathbf{u}_2 + \dots + a_k \mathbf{u}_k.$$

In this case, show that $a_i = \langle \mathbf{v}, \mathbf{u}_i \rangle / \langle \mathbf{u}_i, \mathbf{u}_i \rangle$ for all i . [Remark: If we also have $\langle \mathbf{u}_i, \mathbf{u}_i \rangle = 1$ for all i then the set S is called *orthonormal*, in which case we have $a_i = \langle \mathbf{v}, \mathbf{u}_i \rangle$.]

(b) Use part (a) to prove that every orthogonal set is an independent set.

2. Let W be a vector space and let $U, V \subseteq W$ be subspaces.

(a) Prove that $U \cap V$ is a subspace, but $U \cup V$ need not be a subspace.

(b) Define the sum of subspaces by

$$U + V = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in U, \mathbf{v} \in V\}.$$

Prove that $U + V$ is a subspace of W .

(c) Suppose that $U \cap V = \{\mathbf{0}\}$. In this case, show that for all vectors $\mathbf{w} \in U + V$ there exists a **unique** expression $\mathbf{w} = \mathbf{u} + \mathbf{v}$ with $\mathbf{u} \in U$ and $\mathbf{v} \in V$. [Hint: Suppose that $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$ with $\mathbf{u}, \mathbf{u}' \in U$ and $\mathbf{v}, \mathbf{v}' \in V$. Then ...]

(d) If $W = U + V$ and $U \cap V = \{\mathbf{0}\}$ then we say that W is the *direct sum* of U and V , and we write

$$W = U \oplus V.$$

In this case, show that $\dim W = \dim U + \dim V$. [Hint: Let $\mathbf{u}_1, \dots, \mathbf{u}_m$ be a basis of U and let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be a basis of V . Prove that $\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n$ is a basis of W .]

3. Let V be a real inner product space and let $U \subseteq V$ be a subspace. We define the *orthogonal complement* of U as follows:

$$U^\perp := \{\mathbf{v} \in V : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for all } \mathbf{u} \in U\}.$$

(a) Prove that $U^\perp$ is a subspace of V .

(b) Prove that $U \cap U^\perp = \{\mathbf{0}\}$.

(c) If V is finite dimensional, prove that $V = U + U^\perp$. Combining this with (b) gives

$$V = U \oplus U^\perp,$$

and hence $\dim V = \dim U + \dim U^\perp$. [Hint: Suppose that $\dim U = k$ and $\dim V = n$, so that $k \leq n$. From the Gram-Schmidt procedure (see below), it is possible to find an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ for V such that $\mathbf{u}_1, \dots, \mathbf{u}_k$ is a basis for U . Show that $\mathbf{u}_{k+1}, \dots, \mathbf{u}_n$ are in $U^\perp$ and use this to show that every $\mathbf{v} \in V$ can be expressed as $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$.]

4. Let V be a finite-dimensional real inner product space and let $U \subseteq V$ be a subspace. From Problem 3(c) we know that $V = U \oplus U^\perp$. Hence from Problem 2(c) we know that for every vector $\mathbf{x} \in V$ there exists a unique pair of vectors $\mathbf{x}_U \in U$ and $\mathbf{x}_{U^\perp} \in U^\perp$ such that

$$\mathbf{x} = \mathbf{x}_U + \mathbf{x}_{U^\perp}.$$

Define two functions $P_U, P_{U^\perp} : V \rightarrow V$, called the *orthogonal projections* onto U and $U^\perp$, by setting $P_U(\mathbf{x}) := \mathbf{x}_U$ and $P_{U^\perp}(\mathbf{x}) := \mathbf{x}_{U^\perp}$. Let $\mathbf{u}_1, \dots, \mathbf{u}_k$ be an orthogonal basis for U . In this case, show that

$$P_U(\mathbf{x}) = \frac{\langle \mathbf{x}, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 + \frac{\langle \mathbf{x}, \mathbf{u}_2 \rangle}{\langle \mathbf{u}_2, \mathbf{u}_2 \rangle} \mathbf{u}_2 + \frac{\langle \mathbf{x}, \mathbf{u}_k \rangle}{\langle \mathbf{u}_k, \mathbf{u}_k \rangle} \mathbf{u}_k$$

and

$$P_{U^\perp}(\mathbf{x}) = \mathbf{x} - P_U(\mathbf{x}) = \mathbf{x} - \sum_{i=1}^k \frac{\langle \mathbf{x}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i.$$

[Hint: Use Problem 1(a).]

Challenge Problem (Gram-Schmidt Procedure). Let V be a finite-dimensional inner product space over $\mathbb{R}$. The *Gram-Schmidt procedure* is used to convert any basis $\mathbf{v}_1, \dots, \mathbf{v}_n$ into an orthogonal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$. We inductively define

$$\begin{aligned} \mathbf{u}_1 &:= \mathbf{v}_1, \\ \mathbf{u}_2 &:= \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1, \\ \mathbf{u}_3 &:= \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 - \frac{\langle \mathbf{v}_3, \mathbf{u}_2 \rangle}{\langle \mathbf{u}_2, \mathbf{u}_2 \rangle} \mathbf{u}_2, \\ &\text{etc.} \end{aligned}$$

Afterwards, we may obtain an *orthonormal* basis by replacing each $\mathbf{u}_i$ with $\mathbf{u}_i / \|\mathbf{u}_i\|$.

- (a) Use induction to prove that the following statement holds for all $k \geq 1$:

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\} \text{ and } \mathbf{u}_k \neq \mathbf{0}.$$

[Remark: We need $\mathbf{u}_k \neq \mathbf{0}$ so we can divide by $\langle \mathbf{u}_k, \mathbf{u}_k \rangle$ in the definition of $\mathbf{u}_{k+1}$.]

- (b) Use induction to prove that the following statement holds for all $k \geq 1$:

$$\langle \mathbf{u}_i, \mathbf{u}_k \rangle = 0 \quad \text{for all } i < k.$$

- (c) Define the subspace $U_k := \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$. Use Problem 4 to show that

$$\mathbf{u}_{k+1} = P_{U_k^\perp}(\mathbf{v}_{k+1}) \quad \text{for all } k \geq 1.$$

The idea: We create a new vector $\mathbf{u}_{k+1}$ that is perpendicular to the previous vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ by projecting the original $\mathbf{v}_{k+1}$ onto the orthogonal complement $U_k^\perp$. Parts (a) and (b) show that this idea works.

- (d) The Gram-Schmidt procedure can also be applied in infinite-dimensional vector spaces. Consider the function space $L^2[0, 1]$ with inner product

$$\langle f(x), g(x) \rangle = \int_0^1 f(x)g(x) dx.$$

If we apply Gram-Schmidt to the infinite sequence $1, x, x^2, \dots$, then we obtain an infinite sequence of polynomials $p_0(x), p_1(x), p_2(x), \dots$ called the *shifted Legendre polynomials*. Compute the first three Legendre polynomials and check that they are orthogonal with respect to the inner product.

- (e) Let $U = \text{Span}\{1, x, x^2\} \subseteq L^2[0, 1]$. Continuing from part (d), we know that

$$U = \text{Span}\{p_0(x), p_1(x), p_2(x)\}.$$

Use the formula from Problem 4 to compute the projection $f(x) := P_U(x^3)$ onto the subspace U . Draw the graphs of x^3 and $f(x)$ on the same axes.